

Ex. No.: 10(b)
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FIRST FIT

Aim:

To write a C program for implementation memory allocation methods for fixed partition using first fit.

Algorithm:

1. Define the max as 25.
2. Declare the variable frag[max], b[max], f[max], i, j, nb, nf, temp, highest=0, bf[max], ff[max]. 3: Get the number of blocks, files, size of the blocks using for loop.
4. In for loop check bf[j]!=1, if so temp=b[j]-f[i]
5. Check highest

Program Code:

```
#include <stdio.h>
#define MAX 25
int main () {
    int frag[MAX], b[MAX], f[MAX], bf[MAX], ff[MAX];
    int i, j, Nb, nf, temp;
    printf("Enter the number of memory blocks: ");
    scanf("%d", &Nb);
    printf("Enter the size of each memory block: \n");
    for(i=0; i<Nb; i++){
        printf("Block %d: ", i+1);
        scanf("%d", &b[i]);
        bf[i]=0;
        printf("Enter the number of files processes: ");
        scanf("%d", &nf);
        printf("Enter the size of each file: \n");
        for(i=0; i<nf; i++){
            printf("File %d: ", i+1);
            scanf("%d", &f[i]);
        }
    }
}
```



```

for (i=0; i < nb; i++){
    for (j=0; j < nb; j++){
        if (b[j] == 0 && b[j] >= f[i]){
            f[i] = j;
            b[j] = 1;
            frag[i] = b[j] - f[i];
            break;
        }
        if (j == nb){
            f[i] = -1;
        }
    }
    printf("\nFile No. %d File Size %d Block No. %d Block size %d Fragment %d\n", i+1, f[i], b[i], frag[i]);
    for (i=0; i < nb; i++){
        printf("%d\t\t%d\t\t", i+1, f[i]);
        if (f[i] != -1)
            printf("%d\t\t%d\t\t%d\n", f[i]+1, b[f[i]], frag[i]);
        else
            printf("Not Allocated\t\t\t\t\t\n");
    }
}

```



OUTPUT :

The fragment of blocks are

80
15
23
5
20

Process NO	Process - size	Block - NO	Fragment
P ₁	20	1	30
P ₂	30	2	15
P ₃	50	5	20
P ₄	40	4	5
P ₅	10	3	23

Sample Output:

```
Enter the number of blocks:4
Enter the number of files:3

Enter the size of the blocks:-
Block 1:5
Block 2:8
Block 3:4
Block 4:10
Enter the size of the files:-
File 1:1
File 2:4
File 3:7

File_no:      File_size :      Block_no:      Block_size:      Fragment
1             1             1             5             4
2             4             2             8             1
3             7             4             10            3_
```

Result:

using a program the first fit memory allocation
algorithm is implemented
[Signature]